5.6: Defining Sequences Recursively

Definition.
A **recurrence relation** for a sequence $a_0, a_1, a_2, \ldots$ is a formula that relates each term a_k to a certain subset of its predecessors $a_{k-1}, a_{k-2}, \ldots, a_{k-i}$, where i is an integer with $k - i \geq 0$.
If i is a fixed integer, the **initial conditions** for such a recurrence relation specify the values $a_0, a_1, a_2, \ldots, a_{i-1}$.

Example. Define a sequence $c_0, c_1, c_2, \ldots$ recursively as follows: For every integer $k \geq 2$,

$$c_k = c_{k-1} + kc_{k-2} + 1$$
$$c_0 = 1 \text{ and } c_1 = 2$$

Find the first 5 elements of this sequence.

$C_0 = 1$

$C_1 = 2$

$C_2 = C_1 + 2C_0 + 1 = 2 + 2(1) + 1 = 5$

$C_3 = C_2 + 2C_1 + 1 = 5 + 2(2) + 1 = 10$

$C_4 = C_3 + 2C_2 + 1 = 10 + 2(5) + 1 = 21$

Example. The **Fibonacci Numbers** are a sequence commonly attributed to Leonardo of Pisa (a.k.a Fibonacci). He posed the following problem:

> A single pair of rabbits (male and female) is born at the beginning of a year. Assume the following conditions:
>
> 1. Rabbit pairs are not fertile during their first month of life but thereafter give birth to one new male/female pair at the end of every month.
> 2. No rabbits die.
>
> How many rabbits will there be at the end of the year?

End of month:	0	1	2	3	...	11	12
Pairs of rabbits:	1	1	2	3			
Pairs of fertile rabbits:	0	1	1	2			

$\Rightarrow F_{k+1} = F_k + F_{k+1}$

k	F_k
0	1
1	1
2	2
3	3
4	5
5	8
6	13
7	21
8	34
9	55
10	89
11	144
12	233

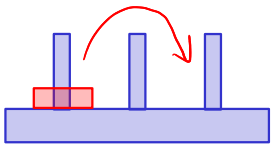
Example. The **Tower of Hanoi** is a puzzle with 8 circular disks of wood with holes in the center. The disks are stacked from the smallest down to the largest. The objective of the game is to move the stack, one disk at a time, from one pole to another without placing a larger disk above a smaller disk:

Tower of Hanoi

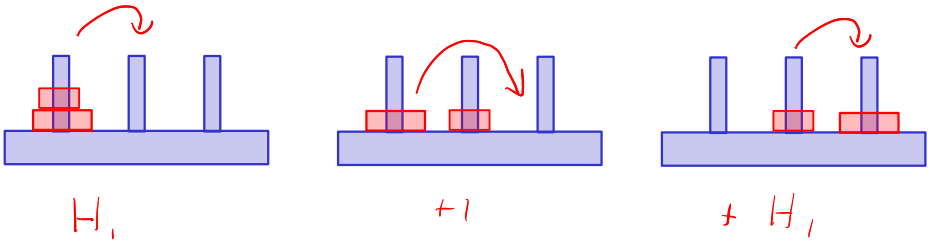
What is the minimum number of moves required to solve this puzzle?

Let H_n be the minimum number of moves to move n disks

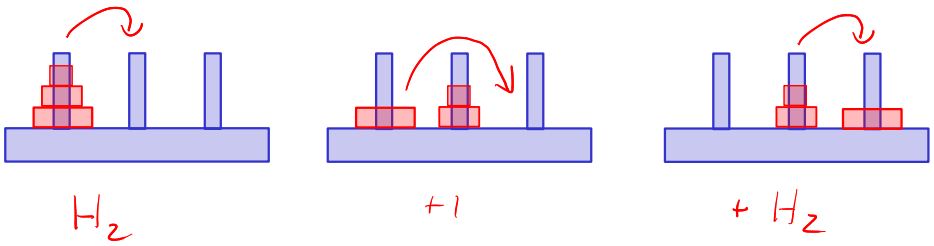
$H_1 = 1$



$H_2 = 2(1) + 1$
$= 3$



$H_3 = 2(3) + 1$
$= 7$



$\vdots$

$$\boxed{H_n = 2H_{n-1} + 1}$$

k	F_k
0	1
1	1
2	2
3	3
4	5
5	8
6	13

Example. Use mathematical induction to prove that for each integer $n \geq 0$,

$$F_{n+2}F_n - F_{n+1}^2 = (-1)^n$$

Base case $(n=0)$

$$F_2 F_0 - F_1^2 = 2 \cdot 1 - 1^2 = 1 = (-1)^0$$

Induction step $(n-1)$

Suppose $F_{n+1} F_{n-1} - F_n^2 = (-1)^{n-1}$

$$\boxed{\text{Def'n: } F_{k+1} = F_k + F_{k-1}}$$

Show it works in general (n)

$$\begin{aligned}
F_{n+2} F_n - F_{n+1}^2 &= (F_{n+1} + F_n) F_n - F_{n+1}^2 && \text{by definition} \\
&= F_{n+1} F_n + F_n^2 - F_{n+1}^2 && \text{distributing} \\
&= F_{n+1}(F_n - F_{n+1}) + F_n^2 && \text{factoring} \\
&= F_{n+1}(-F_{n-1}) + F_n^2 && \text{by definition} \\
&= -(F_{n+1} F_{n-1} - F_n^2) && \text{factoring} \\
&= -1(-1)^{n-1} && \text{by assumption} \\
&= (-1)^n
\end{aligned}$$

Example. Consider the following recursive sequence:

$$x_n = \sqrt{4 + 3x_{n-1}}$$

Find

$$\lim_{n\to\infty} x_n$$

Let $\lim_{n\to\infty} x_n = x^*$

Then $x^* = \sqrt{4+3x^*} \iff (x^*)^2 = 4 + 3x^*$

$\iff (x^*)^2 - 3x^* - 4 = 0$

$\iff (x^*+1)(x^*-4) = 0$

$x^* = -1$ $\boxed{x^* = 4}$

Impossible

$$\boxed{\text{Thus } \lim_{n\to\infty} x_n = 4}$$